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Explained and unexplained tissue loss in corals from the Tropical Eastern Pacific

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ABSTRACT: Coral reefs rival rainforest in biodiversity, but are declining in part because of disease. Tissue loss lesions, a manifestation of disease, are present in dominant *Pocillopora* along the Pacific coast of Mexico. We characterized tissue loss in 7 species of *Pocillopora* from 9 locations (44 sites) spanning southern to northern Mexico. Corals were identified to species, and tissue loss lesions were photographed and classified as those explainable by predation and those that were unexplained. A focal predation study was done concurrently at 3 locations to confirm origin of explained lesions. Of 1054 cases of tissue loss in 7 species of corals, 84% were associated with predation (fish, snails, or seastar) and the remainder were unexplained. Types of tissue loss were not related to coral density; however there was significant geographic heterogeneity in type of lesion; one site in particular (Cabo Pulmo) had the highest prevalence of predator-induced tissue loss (mainly pufferfish predation). Crown-of-thorns starfish, pufferfish, and snails were the most common predators and preferred *P. verrucosa, P. meandrina,* and *P. capitata*, respectively. Of the 9 locations, 4 had unexplained tissue loss with prevalence ranging from 1 to 3% with no species predilection. Unexplained tissue loss was similar to white syndrome (WS) in morphology, indicating additional study is necessary to clarify the cause(s) of the lesions and the potential impacts to dominant corals along the Pacific coast of Mexico.

KEY WORDS: *Pocillopora* · Lesions · Predation · Disease

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INTRODUCTION

Coral reefs rival tropical rainforests as one of the most diverse ecosystems on Earth, occupying less than 0.5% of the planet's surface but harboring 25% of its biodiversity (Hughes 1994, Pandolfi et al. 2003, 2005). Diseases are a major concern in coral reef ecosystem conservation (Richardson et al. 1998) and have been responsible for significant declines of coral reefs in the Western Atlantic (Weil & Rogers 2011). In the Eastern Pacific, we know very little about coral diseases, but efforts are being made to understand the health status of dominant *Pocillopora* species (Rodríguez-Villalobos et al. 2014). A first step in elucidating disease in this region involves morphological descriptions of lesions to assist in explaining the environmental cofactors associated with injuries on coral colonies (Work & Aeby 2006).

Marine invertebrates experience injuries at different frequencies, levels of severity (from minor to lethal events), and from a variety of sources (natural to anthropogenic; Lindsay 2010). Injuries to marine invertebrates are common, and most are repaired; however depending on severity, repair processes are

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energetically costly and can lead to decreased reproduction and growth rates as well as general changes in individual defensive behaviors (Lindsay 2010). Tissue loss is one of the most common lesions observed in coral colonies and is associated with many different reported diseases, including white band, white plague, shut down reaction, skeletal eroding band, and white syndrome (Bythell et al. 2004, Page & Willis 2008, Work & Aeby 2011). Tissue loss lesions are also some of the most insidious for corals, because they have immediate adverse impacts on the colony (cell death), and the causes are often unclear and require detailed laboratory examinations (Aeby 2007, Cervino et al. 2008, Work & Aeby 2011, Miller & Richardson 2012). In contrast, some tissue loss can be explained by predation or intra- or interspecific interactions. Indeed, corallivory can have important ecological manifestations in coral reefs (Rotjan & Lewis 2008) affecting both reproduction and survivorship. Predators can inflict several types of tissue loss lesions in corals, and differentiating these from tissue loss caused by non-predation events (e.g. unexplained tissue loss possibly caused by infectious agents, physiologic disorders, or toxins) can be challenging.

Work & Aeby (2006) emphasize the importance of attempting to differentiate tissue loss lesions in corals that could be explained (e.g. predators) versus those with no grossly identifiable cause (unexplained). This distinction is important, because coral injuries resulting from predation could be used as estimators of reef health status. For instance, fish bites on corals are used as a proxy to reflect corallivore abundance (Jayewardene et al. 2009). Presence of corallivores has also been used to indicate ecosystem and trophic level integrity (Sandin et al. 2008), while their absence can suggest ecological imbalances such as overfishing (Raymundo et al. 2009). Alternatively, unexplained tissue loss in corals might indicate entirely different processes such as disease (Green & Bruckner 2000, Bourne et al. 2015).

Predation can be difficult to document, particularly in the case of predators with skittish behavior such as fish or birds (Rotjan & Lewis 2008, Davis et al. 2012). Observational studies allow one to associate the morphological features of predator-induced injuries (mouth, teeth, jaws) with the predator (Reimchen 1988, Jayewardene et al. 2009). Once lesions induced by a particular predator are systematically described, they can generally be reliably identified during field surveys.

Along the Pacific coast of Mexico, it is common to see lesions in corals caused by fish or invertebrate predation; however these lesions have not been systematically described. In the Eastern Pacific, many species of invertebrates and fish are corallivores (Glynn et al. 1972, Glynn 2004, Gibson et al. 2011) and coral reefs are highly dominated by *Pocillopora* species (Reyes-Bonilla & Calderon-Aguilera 1999). Known invertebrate corallivores in the region include the snails *Jenneria pustulata* and *Coralliophilia monodonta*, and the crown-of-thorns seastar (COTS) *Acanthaster plancii* (Reyes-Bonilla & Calderon-Aguilera 1999, Paz-García et al. 2012, Landa-Jaime et al. 2013). Of known fish corallivores, the pufferfish *Arothron meleagris* is the most abundant predator of *Pocillopora* (Reyes-Bonilla & Calderon-Aguilera 1999). In contrast, butterflyfish, which are common corallivores in the central and western Pacific (Reese 1981, Rotjan & Lewis 2008), are uncommon in the Mexican Pacific, with 5 species documented in low numbers, mainly in the coastal zones, few if any using corals as a major food source and with those that are corallivores being facultative (Rodríguez-Romero et al. 2005, Alvarez-Filip & Reyes-Bonilla 2006, López-Pérez et al. 2013).

Because predator-induced lesions can be a useful proxy of corallivore abundance on reefs, and because these lesions have not been well characterized in the Mexican Pacific, our study had 2 objectives: (1) systematically characterize the tissue loss observed in the dominant cauliflower coral *Pocillopora* to differentiate explained (predation) from unexplained lesions and (2) explore the potential relationships between prey availability and the occurrence of predator-induced injuries.

MATERIALS AND METHODS

Study area

The Pacific coast of Mexico is an oceanographically dynamic region distinguished by wind forcing strongly influenced by topography and 3 major currents including the North Equatorial, Costa Rican Coastal, and California Currents (Wyrtki 1967, Badan-Dangon 1998, Kessler 2006). The coral community distribution is constrained by a narrow continental shelf, scattered suitable habitat, nutrient-rich and turbid waters, and low alkalinity (Badan-Dangon 1998). Our surveys were done in coral communities that span a long latitudinal gradient (8°) and coastline (1800 km; Table 1, Fig. 1). The reefs are dominated by *Pocillopora* spp. (mainly *P. verrucosa*) with rare *Porites* spp. and *Pavona* spp. individuals. The

Table 1. Coordinates, depth range, number of sites per location and area surveyed for *Pocillopora* spp. coral lesions in 9 locations in the Pacific coast of Mexico. Number of transects conducted at each location and date of sampling are also shown

Location	°N	°W	Depth range (m)	Sites (reefs)	Total area surveyed (m^2)	Transects n	Transects Area (m^2)	Date (mo/yr)
La Paz	24.35	110.35	1–6	9	2250	45	50	11/13
Cabo Pulmo	23.41	109.42	2–14	8	1600	40	40	11/10
Isla Isabel	21.85	105.92	2–11	5	1150	23	50	11/10
Islas Marietas	20.70	105.54	2–11	4	900	18	50	3/11
Manzanillo	19.11	104.41	1–8	3	1250	25	50	7/11
Melaque	19.21	104.71	2–11	2	500	10	50	2/12
Ixtapa	17.65	101.62	1–9.6	9	330	33	10	2/14
Papanoa	17.52	101.48	2–5	2	60	6	10	2/14
Huatulco	15.74	96.13	3–7	2	350	7	50	9/11

Mexican tropical Pacific (from 24 to 15° N, Fig. 1) has some of the most important fringing reefs in the region with coral cover ranging from 20 to 50% (Reyes-Bonilla 2003).

Gross description and agents involved in *Pocillopora* tissue loss lesions

We visited 9 locations (representing 44 sites or reefs) along the Pacific coast of Mexico from Oaxaca (south) to Baja California (north). At each site, depending on local and survey conditions, we established 3 to 5 consecutive linear transects ranging in length from 10 to 25 m, laid end-to-end parallel to the coast following depth contours and separated by ca. 5 m (Table 1). Corals within transects were counted within a 2 m swath and identified to species (Veron 2000).

Tissue loss lesions in corals were photographed with a SeaLife DC 1200 camera, quantified as presence/absence, and grossly described as per Work & Aeby (2006) as follows: (1) shape: circular, oblong, irregular, or linear; (2) distribution: focal, multifocal or diffuse; (3) lesion edges: distinct or indistinct; (4) lesion margin: smooth or undulating; and (5) location on the colony: apical, medial or basal. In addition to recording tissue loss in the colonies, we photographed and documented the presence on or near the injured colonies of the invertebrates *Jenneria pustulata*, *Coraliophilia monodonta* and *Acanthaster plancii*, and the pufferfish *Arothron meleagris*. Chaetodontidae were excluded because of low abundances in the region and because these fish do not remove significant amounts of tissues from corals (Rotjan & Lewis 2008).

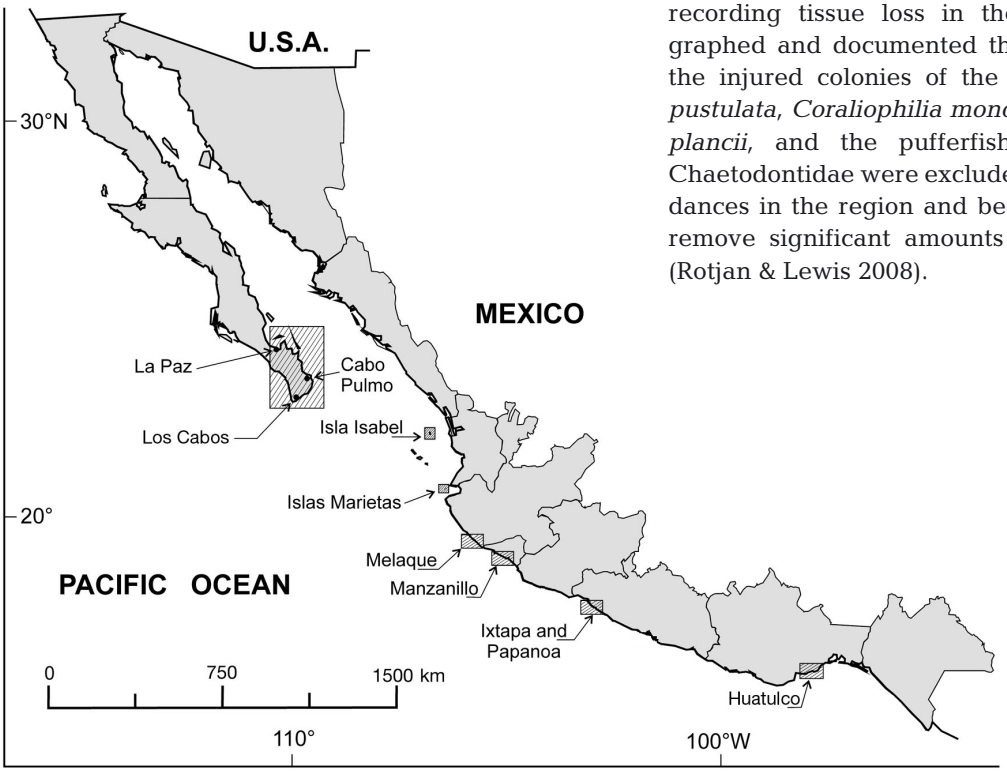

Fig. 1. Locations surveyed for *Pocillopora* spp. coral tissue injuries along the Pacific coast of Mexico

Assessment of predators associated with injuries on colonies

In order to confirm the association between predators and particular morphology of gross lesion, we did focused predation studies at 24 reefs in 3 different locations in Baja California Sur in May and November 2013 (Table 2). At each site, 4 consecutive belt transects (25 × 4 × 2 m for *A. meleagris* and 25 × 2 m for invertebrates) separated by ca. 5 m were established end-to-end parallel to the coastline. Abundance and density of predators (*A. meleagris*, *J. pustulata*, *C. monodonta* and *A. planci*) and coral species were determined along each transect. Furthermore, lesions resulting from tissue loss induced by a particular predator were photographed and characterized. Species identification was done following Kerstich & Bertsch (2006) for invertebrates and Robertson & Allen (2008) for fish. Because their corallivorous behavior could not be reliably assessed using belt transects, we videotaped pufferfish feeding on a healthy *Pocillopora* spp. colony located at one end of each transect and morphologically described the resulting lesion.

Data analysis

Predator diversity was estimated using the Shannon-Wiener index and prey density was calculated as the number of colonies per species per m^2. Overall prevalence of tissue loss by lesion type following Work & Aeby (2006; Table 2) was quantified for each *Pocillopora* species, pooling data from all sites. We tested correlations between coral density and number of types of tissue loss using Spearman's correlation coefficient. A χ^2 test was used to test geographical homogeneity of lesion type versus geographic location. We calculated resource selection ratios (ω_i) with respect to *Pocillopora* species availability (Table S1 in the Supplement at www.int-res.com/articles/suppl/d116p121_supp.pdf) to evaluate potential prey preferences in predators following Schoepf et al. (2010). A confidence interval that encompassed 1, <1 and >1 indicated that the prey was exploited according to, less than, or greater than expected, respectively, from its availability.

Statistical analyses were performed using Statistica ver 7.1 (StatSoft 2005) while diversity indices were assessed through Primer 6 (Clarke & Gorley 2006). Alpha was 0.05.

RESULTS

We surveyed 7940 m^2 representing 44 sites from 9 locations and documented 1042 colonies with tissue loss in *Pocillopora capitata, P. damicornis, P. effusus, P. eydouxi, P. inflata, P. meandrina*, and *P. verrucosa.* Five types of tissue-loss lesions were seen in *Pocillopora* spp. The most common (80%) was an oblong to distinctly circular lesion characterized by eroded skeleton with focal to multifocal distribution on the tips of colony branches. Video recordings confirmed *Arothron meleagris* predation was responsible for this particular lesion in all cases (Fig. 2A). Variably sized irregular diffuse areas of tissue loss revealing intact bare skeleton were associated with *Jenneria pustulata* predation (Fig. 2B). In contrast, *Coralliophilia monodonta* predation (Fig. 2C) resulted in ca. 1 cm oblong (snail footprint) areas of acute tissue loss revealing intact skeleton in different positions along branches. COTS predation (Fig. 2D) was recognized as diffuse tissue loss in a distinct linear pattern extending from the tips to mid-portion of multiple contiguous branches exposing intact skeleton (Table 3). Distribution of tissue loss was mostly focal-to-multifocal (81%), and usually involved skeletal damage (78%) (Table 3).

Unexplained tissue loss comprised 15% of cases (n = 166) and was seen exclusively at Cabo Pulmo, La Paz, Manzanillo, and Ixtapa (Table 3). These were characterized as a diffuse band of sub-acute tissue loss usually at the base of branches with bleached tissue apposed to intact bare intact skeleton progressively overgrown by turf algae with increasing distance from coral tissues (Fig. 3). We saw no predators associated with this type of lesion.

Resource selection ratios revealed predation was directed at different species. *Pocillopora effusus and P. eydouxi* were rarely preyed on relative to species abundance, while *P. meandrina* and *P. capitata* were preferred prey for *A. meleagris* and *C. monodonta*, respectively. *J. pustulata*

Table 2. Coordinates, depth range, and area surveyed for predators on *Pocillopora* spp. coral colonies in 3 locations in the Pacific coast of Mexico during May and November 2013. Number of transects conducted at each location is also shown

Location	°N	°W	Depth range (m)	Area (m^2) Fish	Area (m^2) Invert.	No. of transects Fish	No. of transects Invert.
Cabo Pulmo	23.41	109.42	2–11	6400	3250	64	65
La Paz	24.35	110.35	10–14	4400	2250	44	45
Los Cabos	22.88	109.88	2–14	3300	1700	33	34

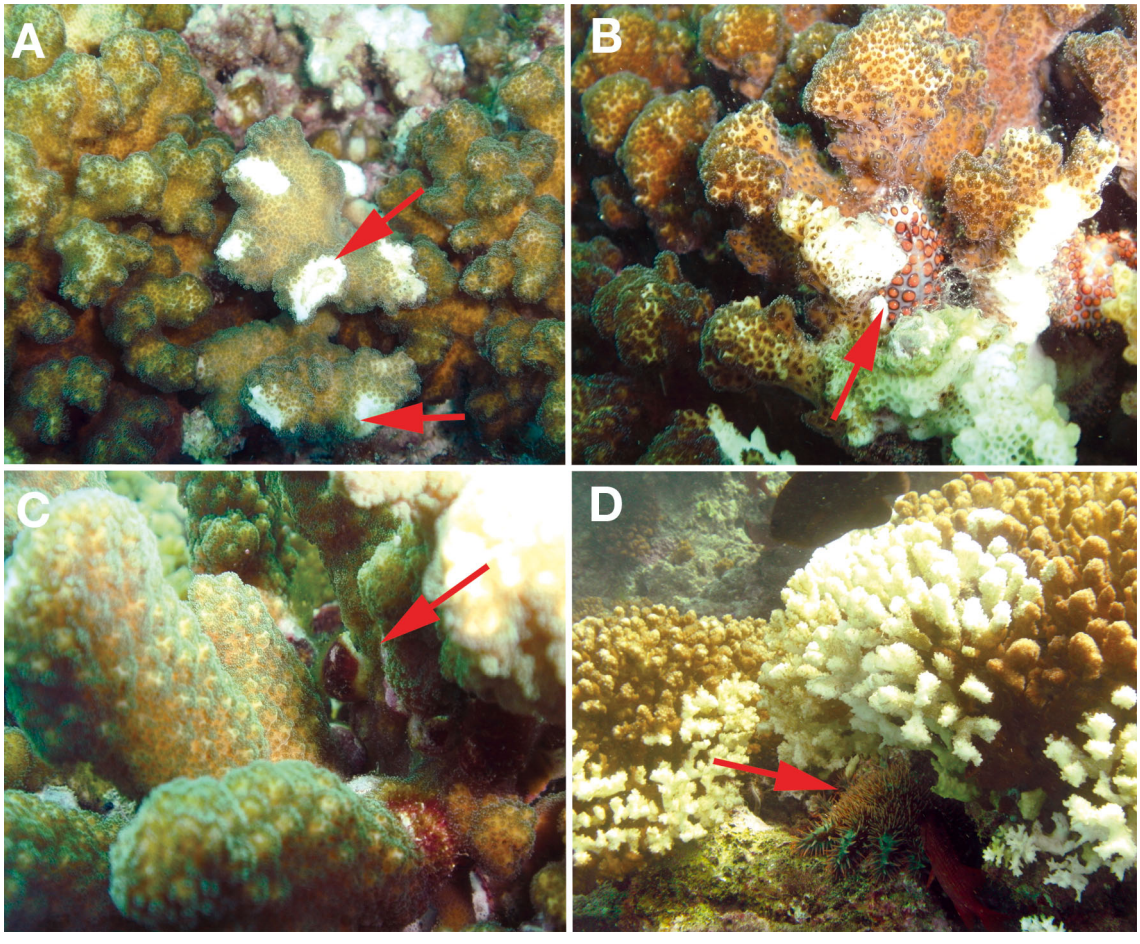

Fig. 2. Characteristics of predation injuries on *Pocillopora* spp. (red arrows, A) and associated predators (red arrows; B,C,D). (A) Pufferfish (*Arothron meleagris*) predation. Multifocal irregular sized distinct areas of acute tissue loss with skeletal erosion generally limited to branch tips. (B) Snail *Jenneria pustulata* predation. Diffuse distinct irregular regions of acute tissue loss on all branch aspects with borders revealing bare white intact skeleton. (C) Snail *Coralliophilia monodonta* predation. Multifocal circular to oblong distinct areas of acute tissue loss on multiple branch sites revealing intact bare skeleton. (D) Crown-of-thorns seastar *Acanthaster plancii* predation. Diffuse acute tissue loss encompassing multiple entire contiguous branches revealing intact bare white skeleton

mainly preyed on *P. verrucosa*, *P. capitata* and *P. damicornis*, which were usually the most abundant corals. All coral species but *P. verrucosa* (preferred prey) remained unused by *Acanthaster plancii*. *Pocillopora damicornis* was only used according to its availability by *J. pustulata*, but was used significantly less than expected by *C. monodonta* and *A. meleagris*, and was unused by *A. plancii*. *P. verrucosa* was the only prey that was preferred by predators or according to its availability ($p < 0.01$; Table 4). Prevalence of tissue loss lesion per species is shown in Table 5.

Cabo Pulmo, La Paz, and Ixtapa had the highest coral densities on the Pacific coast of Mexico (Table 1). There was no correlation between coral species density (Table S2 in the Supplement) and prevalence of lesions in the localities ($r = 0.45$, $p = 0.22$); however we found significant geographic heterogeneity in presence of lesions among sites ($\chi^2 = 623.68$, $p = 0.0$), with Cabo Pulmo and Manzanillo having the highest prevalence of predator-induced tissue loss (Table S3 in the Supplement). All tissue loss types, with the exception of *J. pustulata* predation, were present in La Paz, only pufferfish predation was observed in Isla Isabel and Huatulco, and no predator-induced lesions were detected in Islas Marietas and Papanoa (Table S3 in the Supplement).

At the 3 locations where predators were quantified, *A. meleagris* density was 0.57 ind. m^{-2} at Cabo Pulmo and 0.68 at Los Cabos; *C. monodonta* and *Acanthaster plancii* were only recorded at Cabo Pulmo (0.34 and 0.037 ind. m^{-2}, respectively). None of these predators were recorded at La Paz.

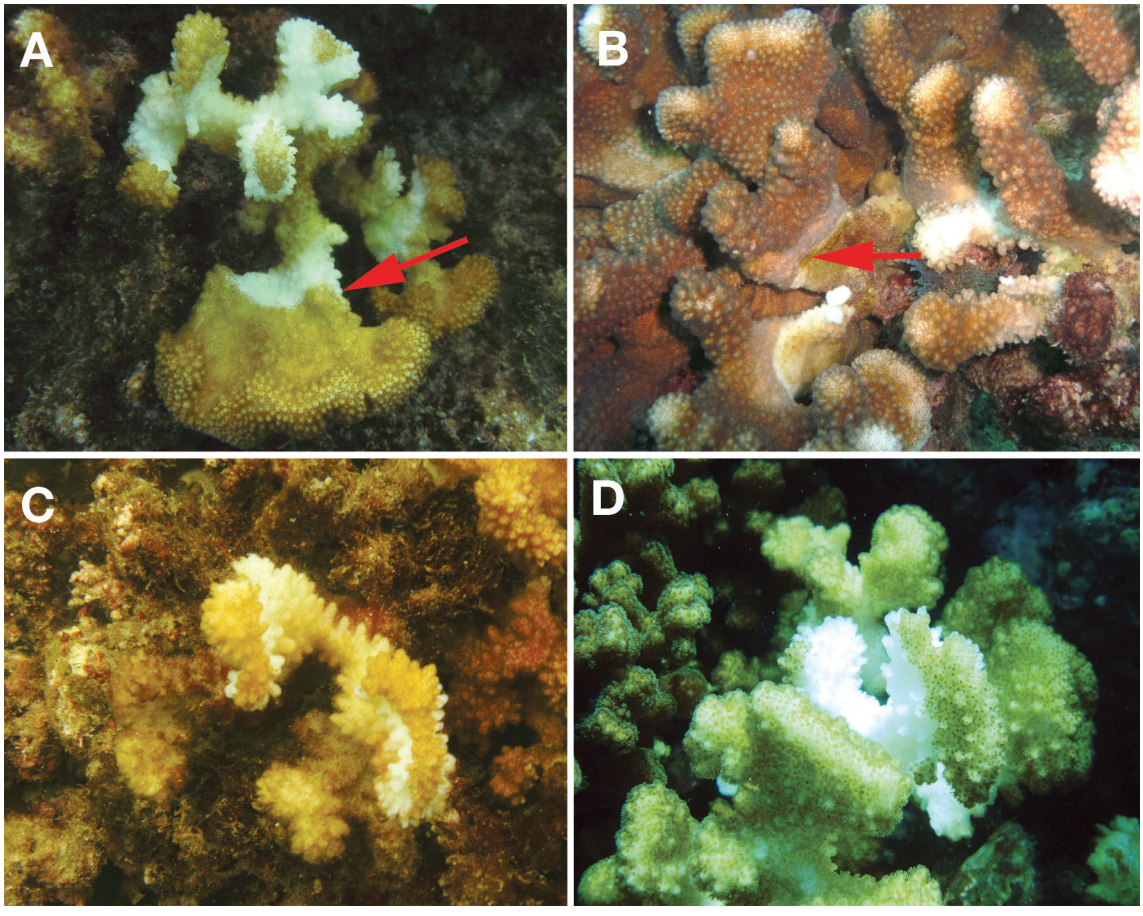

Fig. 3. Unexplained tissue loss in *Pocillopora* spp. Subacute tissue loss characterized by a band of bare intact white skeleton that becomes progressively covered by a patina of turf algae with increasing distance from intact tissues. (A,B) Intact tissue compared with bare skeleton has an indistinct bleached zone (arrow) transitioning to normal tissues; (C,D) the bleached zone is not evident

DISCUSSION

Predation accounted for most of the tissue loss lesions encountered in reefs from the Mexican Pacific, and lesions induced by the pufferfish *Arothron meleagris* accounted for a majority. Pufferfish scrape corals, removing underlying skeleton and leaving characteristic oblong to circular lesions as seen elsewhere (Jayewardene et al. 2009). *A meleagris* is a recognized predator of corals in the Eastern Tropical Pacific (Glynn et al. 1972, 1982, Reyes-Bonilla & Calderon-Aguilera 1999, Palacios et al. 2014) also feeding on algae, crustaceans, and mollusks (Rotjan & Lewis 2008, Moreno et al. 2009, Palacios et al. 2014). However, in the presence of corals, pufferfish seem to prefer *P. meandrina,* like other facultative corallivores such as butterfly fish that also prefer particular coral prey regardless of availability (Pratchett 2005). *A. meleagris* had higher densities in our study than reported in the literature (Reyes-Bonilla & Calderon-Aguilera 1999) for this region (0.57 ind. m^2 to 0.04 ind. m^{-2}).

Documenting pufferfish predation might yield useful insights into coral reef health. The presence of numerous large predators in a marine ecosystem is an accurate measure of optimal ecological function, whereas predator absence indicates ecosystem imbalances (e.g. overfishing) (Friedlander & DeMartini 2002, Friedlander et al. 2008, Sandin et al. 2008). For example, Cabo Pulmo, where we found high prevalence of fish predation, is a protected area in which high prevalence of fish and invertebrate predation could be an indicator of healthy ecosystem represented by more complex food chains (Sandin et al. 2008). In the case of unprotected zones, such as Manzanillo, the presence of increased fish predation on corals could be a consequence of overfishing (Raymundo et al. 2009), that has removed predators of corallivorous fish. Studies that focus on fish community dynamics might help to clarify these processes.

Table 3. Morphological characteristics and occurrence (total number of lesions, n) of the different types of tissue-loss lesions observed on *Pocillopora* spp. colonies. UTL: unexplained tissue loss

Morphology	Predator: Pufferfish *A. meleagris*	Predator: Snails *C. monodonta*	Predator: Snails *J. pustulata*	Predator: Seastar *A. plancii*	UTL
Shape					
Circular	X				
Oblong	X	X			
Irregular			X		
Linear				X	X
Distribution					
Focal	X	X			
Multifocal	X	X			
Diffuse			X	X	X
Edges					
Distinct		X			X
Indistinct	X		X	X	
Margin					
Smooth	X	X	X	X	X
Undulating					
Localization on colonies					
Apical	X			X	
Medial			X	X	
Basal		X		X	
Localization on branches					
Base		X			X
Medial			X		
Tip	X			X	
n	836	29	10	1	166

Table 4. Coral preference of predator as estimated from resource selection ratio ω_i and Bonferroni-corrected 95 and 99% confidence intervals. **Preferred prey species at $p \leq 0.01$; (–) used significantly less than expected from its availability (avoided, $p < 0.05$); UA: used according to availability ($p > 0.05$); UN: unused

Predator	*Pocillopora* sp.	ω_i	Colonies injured (%)
Coralliophilia monodonta	*P. capitata*	**	2.33
	P. damicornis	–	0.15
	P. effusus	UN	0
	P. eydouxi	UN	0
	P. inflata	UA	0.88
	P. meandrina	UA	0.31
	P. verrucosa	UA	0.36
Jenneria pustulata	*P. verrucosa*	UA	0.18
	P. capitata	UA	0.52
	P. damicornis	UA	0.07
	P. effusus	UN	0
	P. eydouxi	UN	0
	P. inflata	UN	0
	P. meandrina	UN	0
Arothron meleagris	*P. meandrina*	**	28
	P. damicornis	–	4.03
	P. inflata	–	4.43
	P. verrucosa	–	11.28
	P. capitata	UA	12.8
	P. effusus	UA	29.41
	P. eydouxi	UA	9.2
Acanthaster plancii	*P. verrucosa*	**	0.02
	P. capitata	UN	0
	P. damicornis	UN	0
	P. effusus	UN	0
	P. eydouxi	UN	0
	P. inflata	UN	0
	P. meandrina	UN	0

We did not find evidence of butterflyfish (Chaetodontidae) predation on corals at our study sites. Butterflyfish remove individual polyps without damaging underlying skeleton (Rotjan & Lewis 2008) leaving small multifocal circular lesions; none of the lesions found in our study fit this description. About half of corallivorous fish species are butterflyfishes (Rotjan & Lewis 2008). In the Tropical Eastern Pacific, there are 10 species of butterflyfish of which 5 exist in Mexico, and of these, only one (*C. meyeri*) is a corallivore (facultative) (Rotjan & Lewis 2008).

Invertebrate predation on *Pocillopora* was associated with *Jenneria pustulata* and *Coralliophilia monodonta*. Irregular diffuse variably sized areas of tissue loss revealing intact bare skeleton characterized lesions associated with *J. pustulata* predation on *Pocillopora* colonies. Our findings were inconsistent with other studies (Navas-Camacho et al. 2010) where lesions attributed to snail predation were more similar to our unexplained tissue loss. Predation by *J. pustulata* on *Porites panamensis* leaves a scar with irregular shape that presents a diffuse dis-

Table 5. Percent of tissue loss lesions explained by predation and unexplained (UTL) in *Pocillopora* species (n = number of colonies) from the Pacific coast of Mexico

Pocillopora sp.	n	*Coralliophilia monodonta*	*Jenneria pustulata*	*Arothron meleagris*	*Acanthaster plancii*	UTL
P. capitata	383	2.33	0.52	12.8	0	3.89
P. damicornis	1363	0.15	0.07	4.03	0	3.52
P. effusus	17	0	0	29.4	0	5.9
P. eydouxi	87	0	0	9.2	0	1.15
P. inflata	113	0.89	0	4.43	0	3.5
P. meandrina	961	0.3	0	28.7	0	1.45
P. verrucosa	3880	0.36	0.18	11.28	0.03	2.13

tribution on colonies (Paz-García et al. 2012), characteristics very similar to those described by us on *Pocillopora* spp. We found that *J. pustulata* does not have a preference for particular *Pocillopora* spp. but eats mainly *P. verrucosa, P. capitata* and *P. damicornis* whereas it avoids *P. effusus, P. eydouxi, P. inflata* and *P. meandrina*. Preference of *J. pustulata* for *Pocillopora* spp. is well known in the Eastern Pacific (Glynn et al. 1982), and in some areas, the snail may be an obligate feeder on this coral genus. For example, in the southeastern Pacific, snails died after after experimental removal of *Pocillopora* spp. (Glynn et al. 1985), whereas in the Gulf of California *J. pustulata* will consume *P. panamensis* in the absence of *Pocillopora* spp. (Paz-García et al. 2012). This snail is highly abundant in the southeastern Pacific where it can reach densities of up to 24 ind. m^{-2} (Glynn et al. 1982), contrasting with our findings where densities were relatively lower.

Coralliophilia spp. are also snail predators of many coral species and are always seen in association with tissue loss (Kružić et al. 2013). *Coralliophilia monodonta* predation on *Pocillopora* spp. manifested as multifocal distinct areas of acute tissue loss revealing intact bare skeleton. The snail leaves a scar on *Pocillopora* colonies underneath its foot (Robertson 1970). In massive *Cladocora caespitosa,* predation by the snail *C. meyenderfii* is characterized by areas of denuded coral skeleton, with single polyps or areas of live tissue found often within the edge of the scar (Kružić et al. 2013). Guzmán-Espinal (1988) reported higher numbers of the snail *C. monodonta* on *Pocillopora* spp. in other localities of the Eastern Tropical Pacific, with a mean number of 7.6 individuals per colony. Based on the reported low predator densities, *C. monodonta* does not appear to contribute significantly to coral mortality at our study sites. It is possible that some snail predation was overlooked because of its cryptic location on the base of *Pocillopora* branches (Guzmán-Espinal 1988, Solis-Bautista et al. 2004).

Acanthaster plancii is one of the most widely recognized predators of scleractinian corals (Kayal et al. 2012, Baird et al. 2013), and its predation can lead to shifts from dominance by branching species such as *Acropora* or *Pocillopora* to massive species, such as *Porites* (Pratchett 2007, Rotjan & Lewis 2008). In our study, COTS predation was easily recognized by presence of the animal in combination with the characteristic scar left from the animal's eversion of its stomach across the coral surface. We found that *P. verrucosa* is the preferred prey for COTS along the Pacific coast of Mexico. While many studies exist on prey selectivity at the genus level by COTS (Pratchett 2001, Kayal et al. 2011, Tokeshi & Daud 2011), most do not include preferences analysis at the species level. Tokeshi & Daud (2011) reported a high preference of COTS for *P. damicornis* whereas they avoided *P. verrucosa*, in contrast to what we found in our study. Many factors such as nutritional value, size, presence of crabs such as *Trapezia* sp., or aversive compounds produced by symbionts could modulate the selectivity of corals by their predators (Cole et al. 2009, Brooker et al. 2013, 2014, Tokeshi & Daud 2011). Likewise, animals could prefer some species over others for such reasons as accessibility or 'innate' choices (Tokeshi & Daud 2011). Outbreaks of COTS globally are a serious problem for reefs, but densities in Los Cabos, Cabo Pulmo and La Paz, have consistently been among the lowest reported in the Eastern Pacific (Reyes-Bonilla & Calderon-Aguilera 1999), and outbreaks have not yet been reported.

Fifteen percent of tissue loss lesions were unexplained, but consistent with white syndrome (WS; Willis et al. 2004), a vague term initially applied to tissue loss diseases affecting a wide variety of corals from the Caribbean and Pacific (Willis et al. 2004, Sussman et al. 2008, Haapkylä et al. 2010, Work et al. 2012, Bourne et al. 2015). A white band originating at the base of coral branches typically characterizes WS (Willis et al. 2004, Ainsworth et al. 2007, Aeby et al. 2010, Andersen et al. 2010, Work et al. 2012). One problem encountered with this lesion is the tendency of investigators to infer infectious causation (usually bacterial) without additional evidence at the tissue level to confirm the etiology. Indeed, studies incorporating examination of corals affected by WS at the microscopic level have reported multiple potential etiologies and host responses (Work et al. 2012). Histology of WS in *Pocillopora* from the Pacific coast of Mexico failed to reveal infectious agents associated with the lesion but rather revealed atrophy of tissues (Rodríguez-Villalobos et al. 2014). As such, senescence may partly explain non-predator-related tissue loss in *Pocillopora* spp. This genus is one of the few scleractinian genera that has determinate growth (Kinzie & Sarmiento 1986). Cessation of tissue regeneration could be one sign of old age in colonies, as demonstrated for *A. palmata* (Meesters & Bak 1995). Elucidating this process in *Pocillopora* might serve to clarify potential causes of WS in this genus in the Eastern Pacific.

The prevalence of unexplained tissue loss in *Pocillopora* (2–3%) was somewhat higher than WS reported in some locations in the Pacific for other coral species, including *Montipora* spp. (0.23 ± 0.09% SD)

(Aeby et al. 2010) and *Acropora* spp. (0.8%) (Aeby 2006) in the Hawaiian islands, but lower in comparison with *Acropora* spp. corals from America Samoa (8.3–8.8%; Aeby et al. 2011). General species prevalence of WS in reefs from Heron Island in the Great Barrier Reef was as low as 0.5% (Haapkylä et al. 2010). We caution, however, that such comparisons may not have much meaning because it is likely that there are genus-specific variations in host response to WS. Because WS has caused catastrophic declines of corals elsewhere (Aronson & Precht 2001), it behooves us to understand more about its pathogenesis and demographic effects, particularly in reefs dominated by a single coral genus, such as those of the Mexican Pacific.

We did not find an association between density of corals and prevalence of tissue loss. Only predators highly dependent on live coral availability, such as obligate corallivores, are expected to be tightly correlated with changes in coral abundance (Cox 1994, Beldade et al. 2015). In the case of pufferfish, a generalist feeder, although its density and abundance on the Pacific coast of Mexico may not necessarily depend on live coral abundance (Beldade et al. 2015), pufferfish numbers could be higher where its preferred prey is available (Pratchett 2005). The highest density of pufferfish's preferred prey (*P. meandrina*) was recorded in Cabo Pulmo where we also found the highest prevalence of predation by *A. meleagris.* High general coral species density (not specifically *P. meandrina*) found in Ixtapa and Papanoa coincides with low prevalence or absence of fish predation.

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